

PROJECT EXPO – TECHNICAL REPORT

SMART PAWS

GPS-Based Dog Safety Belt with 10 m Geofencing

Project Expo Demonstration Report

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Academic Year: _____ Date: _____

PROJECT DECLARATION

We present SMART PAWS, a GPS-based dog safety belt prototype designed to monitor a defined HOME zone and identify when the wearable moves beyond a 10-meter geofence.

The prototype integrates an ESP32 controller, NEO-6M GPS module, OLED display, buzzer, push button and an Android application connected through a phone Wi-Fi hotspot.

ABSTRACT

SMART PAWS is a low-cost wearable safety prototype intended to help keep a dog within a defined safe area. The NEO-6M provides latitude, longitude and satellite information, while the ESP32 processes the position, stores HOME and determines whether the current position is inside or outside a 10-meter geofence.

When the distance is 10 meters or less, the system reports SAFE. When it exceeds 10 meters, the system reports OUTSIDE and activates a buzzer while updating the OLED. The Android application provides live status and demonstration controls through a phone Wi-Fi hotspot.

A clearly labelled simulation mode is available for indoor exhibition conditions when satellite acquisition is weak.

1. INTRODUCTION

SMART PAWS combines location monitoring with a simple geofence decision in a compact wearable prototype. The ESP32 performs the main processing while the GPS module supplies location data. The OLED and buzzer provide local feedback, and the Android application provides the owner interface.

2. PROBLEM STATEMENT

A dog may move away from a permitted area without the owner immediately noticing. The project focuses on a low-cost prototype that compares current position with a stored HOME point and alerts when the distance exceeds 10 meters.

3. OBJECTIVES

- Acquire latitude, longitude and satellite information using NEO-6M.
- Use ESP32 for GPS parsing and geofence processing.
- Define and store a HOME reference point.
- Implement a 10-meter geofence.
- Display position, distance and SAFE/OUTSIDE on OLED.
- Activate a buzzer beyond 10 meters.
- Connect to a phone hotspot and Android application.
- Provide a clearly labelled indoor simulation fallback.

4. SYSTEM OVERVIEW

The presentation identifies six functional elements: ESP32 controller, NEO-6M GPS, Android app, OLED display, buzzer and push button. The flow is GPS position → ESP32 processing → geofence decision → OLED/buzzer alert → Android status.

Component	Primary Role
ESP32	GPS processing, geofence decision, Wi-Fi communication
NEO-6M GPS	Latitude, longitude, satellite count
OLED	SAFE/OUTSIDE and distance
Buzzer	Alert when distance > 10 m
Push Button	HOME calibration trigger
Android App	Live status, distance and demo control

5. HARDWARE COMPONENTS

- ESP32 development board — main controller and Wi-Fi interface.
- NEO-6M GPS module — position source.
- 0.96-inch OLED — local display.
- Active buzzer — audible warning.
- Push button — HOME trigger.
- Jumper wires/breadboard — prototype interconnections.
- USB power/power bank — convenient demonstration power.

6. HARDWARE WIRING AND PIN MAPPING

Component	Pin	ESP32 Pin	Function
NEO-6M GPS	TX	GPIO16 (RX2)	Receives GPS data
NEO-6M GPS	RX	GPIO17 (TX2)	GPS configuration
OLED	SDA/SCL	GPIO21/GPIO22	I2C display
Buzzer	+	GPIO25	Alert output
Push Button	Signal	GPIO27	HOME trigger
All modules	VCC/GND	3V3/GND	Shared power/ground

7. WORKING PRINCIPLE

- Acquire — NEO-6M provides location and satellite data.
- Process — ESP32 parses GPS and calculates distance from HOME.
- Decide — ≤ 10 m is SAFE; > 10 m is OUTSIDE.
- Alert — OLED updates and buzzer sounds in OUTSIDE.
- Report — status is provided to the Android app over the hotspot.

- Indoor fallback — labelled simulation demonstrates the same logic when GPS lock is weak.

8. GEOFENCE LOGIC

HOME is the reference point. The presentation specifies Haversine distance for geographic separation. The boundary is fixed at 10 meters for the prototype.

Condition	State	Alert
Distance $\leq$ 10 m	SAFE	OLED SAFE; buzzer silent; app SAFE
Distance $>$ 10 m	OUTSIDE	OLED OUTSIDE; buzzer ON; app warning

9. ANDROID APP AND PHONE HOTSPOT

The phone acts as the local Wi-Fi access point. The ESP32 joins the hotspot and runs an HTTP server on port 80. The Android app uses the ESP32 local IP to read status and control the demonstration simulation.

Phone	ESP32	Android App
Enable hotspot	Join hotspot	Enter ESP32 IP
Same local network	HTTP server :80	Test connection
Control demo	Update state	Show SAFE / OUTSIDE
Owner interface	Provide status	Display distance and state

10. TESTING AND DEMONSTRATION PROCEDURE

- Power ON — ESP32 boots and OLED initializes.
- GPS/DEMO — use live GPS or labelled simulation coordinates.
- Set HOME — establish the reference point.
- SAFE TEST — verify SAFE at ≤ 10 m.
- OUTSIDE TEST — verify OUTSIDE and buzzer at > 10 m.
- RETURN TEST — return inside the boundary and verify SAFE.

11. RESULTS AND OBSERVATIONS

Test Area	Observation
GPS	Live coordinates and satellite information are supported; simulation is available when satellite lock is weak.
SAFE	System reports SAFE at or below 10 m.
OUTSIDE	System reports OUTSIDE and activates buzzer beyond 10 m.
OLED	Displays status/distance information.
Android	Receives ESP32 status through the phone hotspot.
Simulation	Reproduces the same 10 m state transitions for indoor demo.

12. ADVANTAGES

- Low-cost hardware approach.
- No subscription for the local prototype.
- Real-time OLED and buzzer alerts.
- Simple configurable geofence.
- Compact wearable concept.
- Android owner interface.
- Phone hotspot avoids a separate router.

13. APPLICATIONS

- Home and garden pet safety monitoring.
- Farm and field animal boundaries.
- Wearable safety systems after suitable redesign and validation.
- Campus, warehouse and local asset-boundary monitoring.

14. LIMITATIONS AND DEMO CONSIDERATIONS

- GPS satellite acquisition depends on the environment and is generally easier outdoors.
- A 10-meter boundary can be sensitive to normal GPS position variation.
- The hotspot architecture is local and is not equivalent to worldwide cellular tracking.
- The indoor simulation is clearly labelled and demonstrates the geofence/alert workflow without claiming simulated coordinates are live GPS.

15. FUTURE ENHANCEMENTS

- Li-Po battery integration and low-power modes.
- Cloud logging with SMS/push alerts.
- Multiple geofence zones with adjustable radii.
- Bluetooth short-range fallback.
- Improved wearable enclosure and weather protection.
- GPS filtering to reduce false boundary transitions.

16. CONCLUSION

SMART PAWS demonstrates an end-to-end geofence safety workflow using ESP32, NEO-6M GPS, OLED, buzzer, HOME calibration and an Android application. The prototype turns location into a simple safety decision: SAFE within 10 meters and OUTSIDE beyond 10 meters. The simulation mode provides a practical indoor exhibition fallback while keeping the same geofence and alert behaviour demonstrable.

17. TEAM INFORMATION

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SMART PAWS — GPS decision logic → local alerts → Android status